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# hexmb: a structured O-grid hexahedral mesh engine for morphing cardiovascular geometries

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**Software**

- Review
- Repository
- Archive

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## Summary

hexmb builds all-hexahedral **O-grid (butterfly)** meshes and writes them directly as an OpenFOAM `constant/polyMesh` (Weller et al., 1998). An O-grid fills each cross-section with a square "H" core block surrounded by four boundary-fitted petal blocks: all cells are hexahedra, the boundary is honoured, and there is no point where the mesh converges to a singular axis. Because of that, a single O-grid mesh can be morphed through large deformation — such as a left ventricle contracting by more than half its volume every heartbeat — as one iso-topological mesh, with no remeshing between phases.

The engine is one operation applied to different inputs: sweeping a butterfly cross-section along a stack of rings builds a straight pipe, a bend, a tapered vessel, or a heart chamber. A *telescoping* variant joins coarse apex and valve caps to a fine body through non-conformal interfaces, so the body can carry the cell budget while the closing regions stay coarse enough to survive the contraction. hexmb ships with a reader for public Cardiac Atlas Project left-ventricle finite-element models (Fonseca et al., 2011; Radau et al., 2009), so the whole path from a public, CC0-licensed heart geometry to a checked hexahedral mesh is reproducible with no private data.

## Statement of need

Moving-boundary computational fluid dynamics (CFD) of the cardiac left ventricle (LV) needs a volume mesh that stays valid as the cavity deforms through the beat. The natural structured mesh — rings converging to the apex, like lines of longitude on a globe — has a polar-axis singularity: cells at the apex have almost no volume and invert once the ventricle contracts. Cut-cell meshers such as OpenFOAM's `snappyHexMesh` avoid the singularity but need fragile per-phase remeshing and give up the near-wall hexahedral layers that gradient-based quantities (for example wall shear stress) depend on. O-grid / butterfly topologies are standard in structured grid generation (Thompson et al., 1999), but general, open tooling that *constructs and morphs* them for a deforming biological chamber and exports straight to OpenFOAM has not been readily available.

hexmb fills that gap. Its numpy-only core assembles the block topology once, deduplicates shared nodes by union-find, gates each cell on the scaled Jacobian (Knupp, 2001), writes a valid OpenFOAM `polyMesh`, and morphs the mesh across the cycle by radial-basis-function interpolation of the moving wall (Boer et al., 2007). The telescoping variant couples its coarse and fine regions through the non-conformal arbitrary mesh interface (Farrell & Maddison, 2011), the same interpolation OpenFOAM's coupling is built on. The software targets the growing interest in image-driven, prescribed-motion LV hemodynamics (Peighambari et al., 2026) by supplying the mesh half of that pipeline as open, tested, reproducible software, and it is written to generalise beyond the heart to any single-lumen, star-shaped, genus-0 geometry (pipes, bends, arteries, single chambers).

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## Functionality

- **Templates**: pipe, elbow, single-lumen STL tube, single LV chamber, and a telescoping multi-resolution LV.
- **Direct OpenFOAM export**: `constant/polyMesh` with patch tagging (`wall` / `inlet` / `outlet`), no intermediate format.
- **Quality and verification**: per-cell scaled Jacobian gate, plus a `checkMesh`-based `verify` command.
- **Data input**: a reader for Cardiac Atlas Project `.exnode` LV models and an STL surface exporter, so a public heart geometry can be meshed end to end.
- **Interfaces**: a command-line tool (`hexmb pipe|elbow|cap|lv|stl|verify`) and a small Python API.

The repository includes a worked example (`scripts/cap_to_mesh.py`), an `examples/` directory with a ready-to-open pipe case and a CAP endocardial surface, and a self-contained test suite.

## Limitations

`hexmb` applies to a single connected lumen describable as a stack of star-shaped cross-sections along a centreline; branches, junctions, and multi-lumen trees are out of scope. Assembling the telescoping non-conformal coupling requires OpenFOAM; the numpy-only build and quality gate do not.

## Acknowledgements

The left-ventricle demonstration uses the Cardiac Atlas Project's Sunnybrook Cardiac Data, released into the public domain (CC0) (Fonseca et al., 2011; Radau et al., 2009).

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